1. Isogeny-kernel vs μ_k factor bases for $m = 2$ decomposition

Runs 2026-11-02, from 01:39Z. Branch `claude/isogeny-kernel-factor-base-20261102`. Protocols are frozen at `de0b5628` (`PROTOCOL.md`, Q1–Q3) and `e2cb7331` (`PROTOCOL_Q4_SEARCH.md`, Q4). The task cards CARD-1 and CARD-2 were executed by the mechanical tier and assessed by the director (`cards/`).

Scope. This is a component study at $m = 2$ and toy primes ($p \leq 40961$). It looks at how a factor base is **presented** to a Gröbner solver. There is no ECDLP cost claim and no rho row. The distance remaining is stated against the μ_k reference.

Instrument. The engine is a native Rust Buchberger (normal strategy, both Buchberger criteria) identical for every arm. The primary unit is `termops`, a deterministic count: one per coefficient multiply-subtract on a reductor term or an S-polynomial term. Wall time is recorded but unadmitted, because the host was a shared cloud container with no isolation tier. The EKS route was not reachable from the session (no AWS identity), so both runs were local. Every primary column is host-independent, and a replay elsewhere must reproduce it byte for byte. Gates: 6144 + 6828 systems, with **0** brute-force/evaluation gate failures.

1.1. Results at a glance

Q1 yield	**Null in 15/15 cells** (frozen verdict `null_H1_0_retained`). The median normalised yield r of the isogeny V minus that of a random V lies between -0.020 and $+0.029$, with $\lvert\Delta_r^{\perp}\rvert \leq 3.3\%$. The positive control is valid in every cell (r_{PC} = 0.06–0.49 vs r_R = 0.87–1.02). The user's 7/8 vs 3/8 was noise.
Q2 cost	The user's slowdown reproduces: lex B/A_0 = 6.4–11.7 in 16/16 cells. The frozen H2 verdict is **inconclusive**: the rule held in 8/16 grevlex comparisons, with 0 falsifying. All 8 misses are in the "nodec" stratum, where B/A_1 = 1.58–1.77 falls under the pre-set 2× threshold. Descriptively, Möbius-presented μ_k (A2) costs 0.97–1.00× the Vélu chain (B) in every grevlex and lex cell. The whole gap follows the presentation, not the set V.
S1 density	A0 and A1 have exactly 1.000 nonzeros per product. A2 has 3.11 ($k = 3$) and 5.44 ($k = 4$); B has 3.27 and 5.75. Prediction holds.
Q3 terminal	With the order-4 terminal, B is radical, with $\lvert V\rvert = 2^k$ in k steps. The user's 2-torsion terminal (B2) is non-radical: 2^k distinct points of total multiplicity 2^{k+1} in $k + 1$ steps. B2 costs 7.3–12.9× B in grevlex and 5.3–44× in lex. At $k = 4$ the lex values are lower bounds: all 384 of those systems were censored at the budget, and no other Q2 or Q4 row was censored. Prediction holds.
Q4 lex	**Improvement, 16/16** (frozen verdict). Coordinates selected on a disjoint split cut termops to **0.21–0.30**× native Vélu on held-out targets. The gap to μ_k shrinks from 6.4–11.7× to **1.9–2.6**×.
Q4 grevlex	**No improvement, 0/16**: selected representation / native = 0.82–1.17. The S1-optimal coordinates are 0.92–1.24×. The inverse-variable form is worse (1.04–1.87×). The gap to μ_k stays at 1.5–2.2×.
Kernels	Flat. Native density across 40+ kernels per cell ranges over 3.20–3.27 ($k = 3$) and 5.66–5.75 ($k = 4$), and the best-density kernel costs 0.98–1.01× native. Four $j = 0$ kernels (found only at $p = 12289$) match the random kernels. No $j = 1728$ curve with a point of order 2^{k+2} was found.
Obstruction	None of the 11811 valid Vélu representations reached density 1 (minimum 1.359), as the ramification-alignment argument predicts (§4).
Dickson	The x-only $\mathbb{G}_m$ chain $t' = t^2 - 2$ costs 0.98–1.03× μ_k in both orders, despite density 1.69/2.33. So working only on the x-line is not what costs; the cost is elliptic.

1.2. Q1: yield

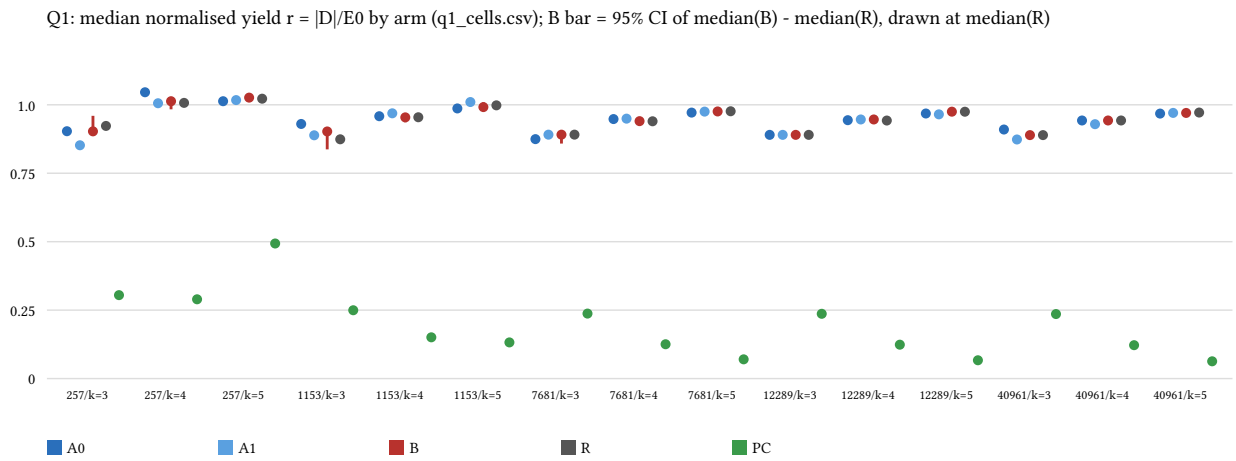


Figure 1: Median normalised yield $r = |D|/E_0$ per arm and cell (`q1_cells.csv`). The red bar is the 95% bootstrap CI of median(r_B) − median(r_R), drawn at median(r_R).

1.3. Q2 and S1: presentation explains the cost

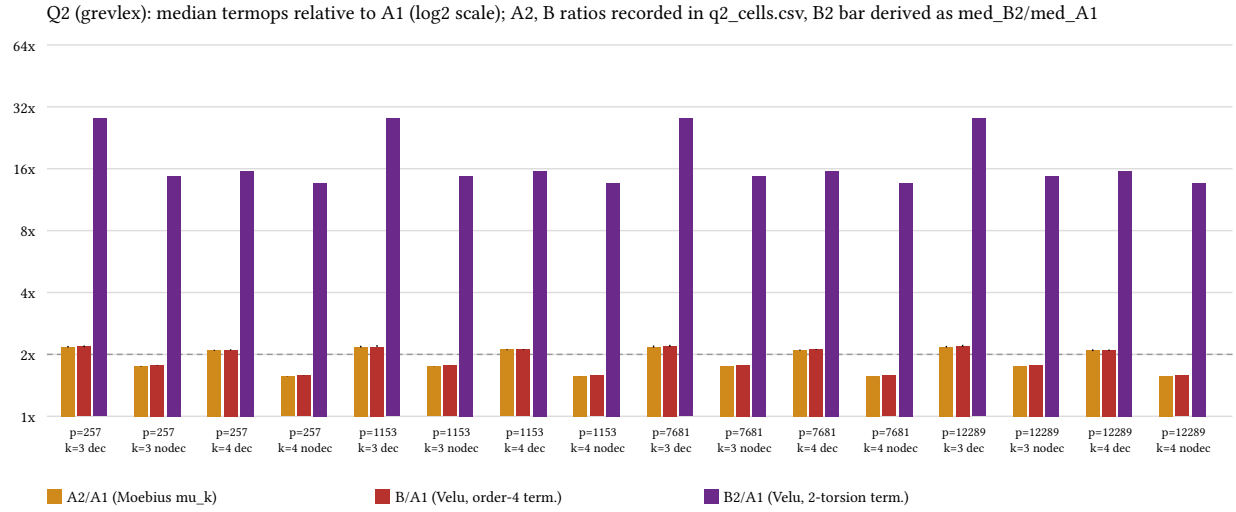


Figure 2: Grevlex: median termops relative to A1 (`q2_cells.csv`). The dashed line is the pre-registered 2× threshold.

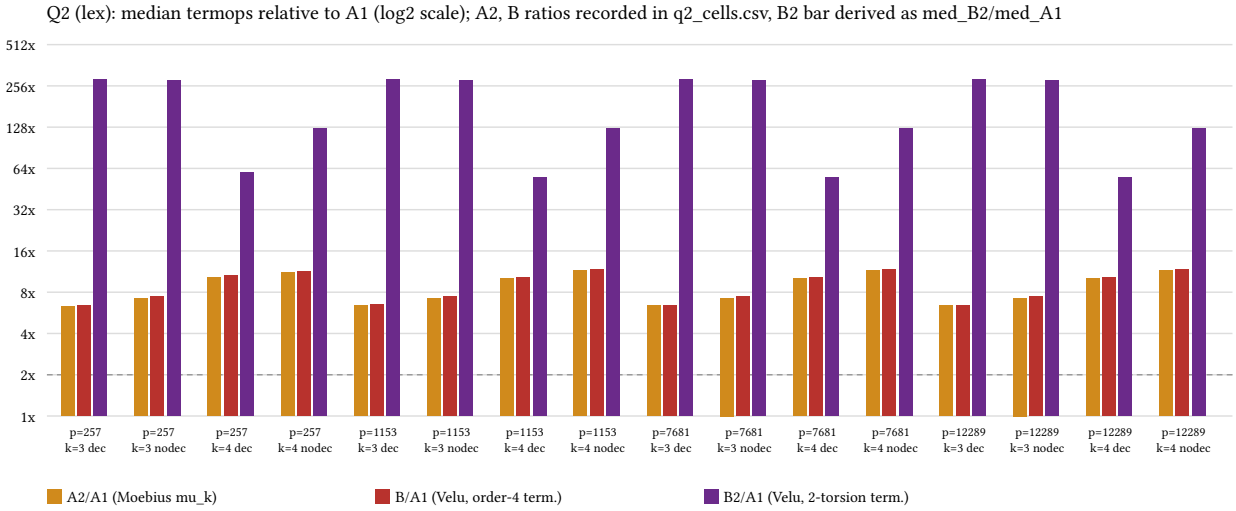


Figure 3: Lex, the user's order. At $k = 4$, every lex B2 system hit the $2 \cdot 10^7$ termops budget (censored), so those B2 bars are lower bounds.

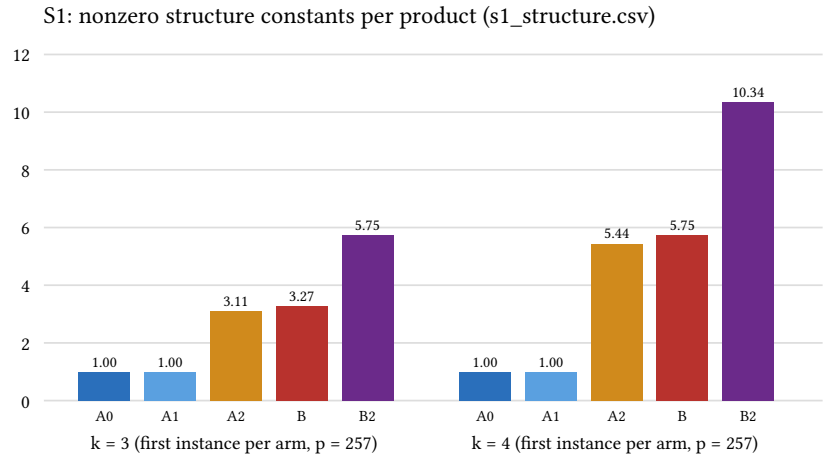


Figure 4: Nonzero structure constants per product in the chain quotient algebra (`s1_structure.csv`).

1.4. Q4: the representation search

Why density 1 is out of reach. A step is monomial ($u' = \lambda u^2$) in suitable coordinates only when its two ramification points sit at 0 and ∞ and its branch points are the next step's ramification points. For a 2-isogeny x-map with kernel $\langle T \rangle$:

- the ramification points are $x\left(\frac{1}{2}T\right)$, which are 4-torsion;
- the branch points are 2-torsion abscissas of the codomain.

The next step ramifies at the halves of its own kernel point, again 4-torsion, so the alignment always fails. For $\ell \geq 3$ the x-map has ramification index ≤ 2, so a monomial step is impossible. This is a written argument with an empirical check (0 of 11811); a Lean proof is follow-up work.

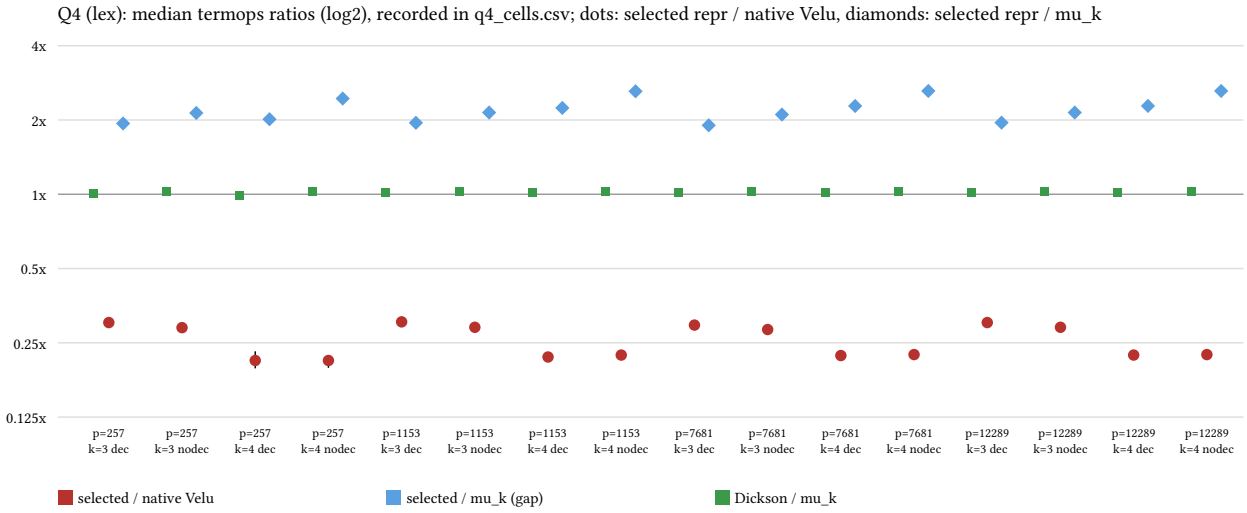


Figure 5: Lex: selected representation / native Vélu (dots, 95% CI) and selected / μ_k (diamonds; the remaining gap), with Dickson / μ_k (squares). Source: `q4_cells.csv`.

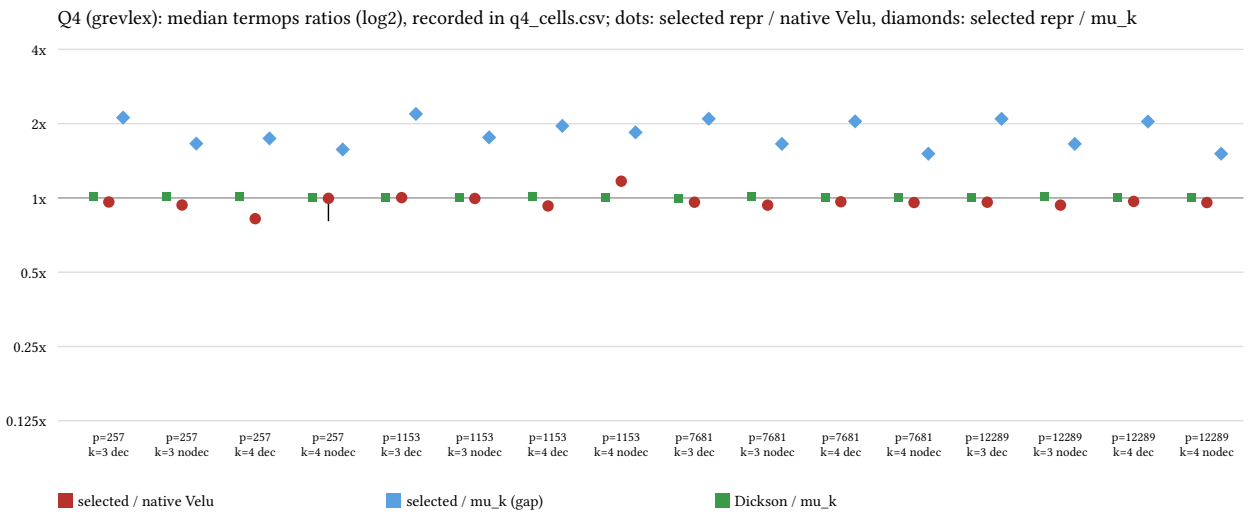


Figure 6: Grevlex, same layout.

The selected lex coordinates are uniform recipes such as `gen>0;ram>inf` (the image of G sent to 0, a ramification point to ∞), or a greedy per-level refinement of one; `q4_candidates.csv` names each one. S1 density is a usable proxy in lex, with Spearman 0.56–0.80 per cell. It is not one in grevlex (median 0.13).

1.5. Caveats

- Toy primes and $m = 2$. Constant factors in a hand-written Buchberger may not carry over to F4/msolve, which this session could not build (external code was blocked).
- The cells at $p = 1153$ were seen in Q4 development before the freeze, and the termops-selection split was added because of what they showed. The frozen run recomputes them, and the other six cells reproduce the same pattern.
- One confirmation target at (7681, 3) has no V abscissa lifting to E ($|F| = 0$), so its yield ratio is NaN, as recorded. With D empty, no "dec" targets can be drawn there, so `q4_systems.csv` has 6828 rows rather than 6912. The 84 missing rows are the 7 Vélu candidates × 6 × 2 orders at that target, so those cells' "dec" medians rest on 3 targets. The director found the shortfall and the code explains it (`eval_systems` draws "dec" targets only when $D \neq \varnothing$).
- `host.txt dirty 2` counts untracked paths, because the `run.sh` filter does not match root-relative porcelain paths. All bound files hash to their card values.
- The relay check produced a false negative caused by output order in `tools/task_card.py` (`cards/relay/CARD-1.relay-recheck.md`).

1.6. Raw records

`results/run-20261102/` and `results/run-q4-20261102/`: CSVs, `verdicts.json`, `q4_verdicts.json`, `SHA256SUMS`, `host.txt`, and the receipts beside them.